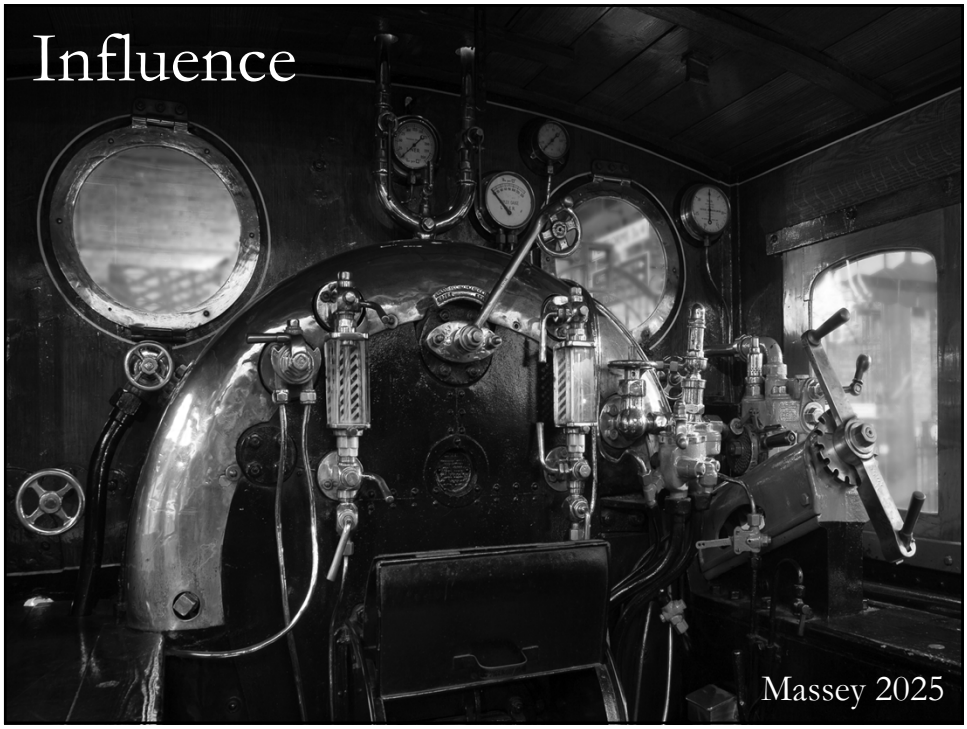


Course Admin (1 of 2)

Leverage Inventory

- Evaluator assessments due Thursday night
- Self-assessment due Thursday night

"Triple Helix" Exercise

- 20-minute exercise sometime over next 8 days
- Instructions & groupings posted last Friday
- Results due by 1p Tuesday Nov 18
- 7% of your course grade

Course Admin (2 of 2)

Introductory Videos

- Evaluations are complete
- Reports delivered by email tomorrow (Tue)
- Debrief the exercise next class (Wed)

Next Class: Persuasion II

- Review your video assessment
- Read the "Warmth & Competence" teaching note
- Do the "Lipman Videos" assignment
  - Watch four 3-minute videos from Lipman semifinalists
  - Reply to survey questions about the videos

Rhetoric in research: An interview



General Article

False-Positive Psychology: Undisclosed Flexibility in Data Collection and Analysis Allows Presenting Anything as Significant

Joseph P. Simmons[1], Leif D. Nelson[2], and Uri Simonsohn[1]
[1]The Wharton School, University of Pennsylvania, and [2]Haas School of Business, University of California, Berkeley

Abstract

In this article, we accomplish two things. First, we show that despite empirical psychologists' nominal endorsement of a low rate of false-positive findings (≤ .05), flexibility in data collection, analysis, and reporting dramatically increases actual false-positive rates. In many cases, a researcher is more likely to falsely find evidence that an effect exists than to correctly find evidence that it does not. We present computer simulations and a pair of actual experiments that demonstrate how unacceptably easy it is to accumulate (and report) statistically significant evidence for a false hypothesis. Second, we suggest a simple, low-cost, and straightforwardly effective disclosure-based solution to this problem. The solution involves six concrete requirements for authors and four guidelines for reviewers, all of which impose a minimal burden on the publication process.

Keywords
methodology, motivated reasoning, publication, disclosure

Received 3/17/11; Revision accepted 5/23/11

Our job as scientists is to discover truths about the world. We generate hypotheses, collect data, and examine whether or not the data are consistent with those hypotheses. Although we aspire to be accurate, errors are inevitable.

Perhaps the most costly error is a *false positive*, the incorrect rejection of a null hypothesis. First, once they appear in the literature, false positives are particularly persistent. Because null results have many possible causes, failures to replicate previous findings are never conclusive. Furthermore, because it is uncommon for prestigious journals to publish null findings or exact replications, researchers have little incentive to-even attempt them. Second, false positives waste resources: They inspire investment in fruitless research programs and can lead to ineffective policy changes. Finally, a field known for publishing false positives risks losing its credibility.

In this article, we show that despite the nominal endorsement of a maximum false-positive rate of 5% (i.e., $p \leq .05$), current standards for disclosing details of data collection and analyses make false positives vastly more likely. In fact, it is unacceptably easy to publish "statistically significant" evidence consistent with any hypothesis.

The culprit is a construct we refer to as *researcher degrees of freedom*. In the course of collecting and analyzing data, researchers have many decisions to make: Should more data be collected? Should some observations be excluded? Which conditions should be combined and which ones compared? Which control variables should be considered? Should specific measures be combined or transformed or both?

It is rare, and sometimes impractical, for researchers to make all these decisions beforehand. Rather, it is common (and accepted practice) for researchers to explore various analytic alternatives, to search for a combination that yields "statistical significance," and to then report only what "worked." The problem, of course, is that the likelihood of at least one (of many) analyses producing a falsely positive finding at the 5% level is necessarily greater than 5%.

This exploratory behavior is not the by-product of malicious intent, but rather the result of two factors: (a) ambiguity in how best to make these decisions and (b) the researcher's desire to find a statistically significant result. A large literature documents that people are self-serving in their interpretation

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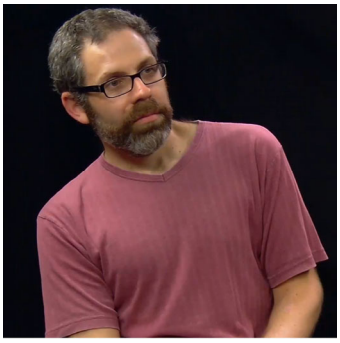
Over 5,000 citations in the 10 years since published in 2011.

Arguably the most influential paper in the "reproducibility crisis" in psychology.

What role did persuasion play?

Simmons & Simonsohn





What role does rhetoric play in science?

Do you want to be effective or expressive?

Advice for fighting effectively, from Professors Simmons & Simonsohn:

"If it feels good to say it, don't."

First, a tale of two economists

May 2000
Finance committee of major research foundation
Question: Reduce equity holdings (from 90%)?

Two eminent economists, both board members, recommend major reductions (e.g., 0%).

- One economist is in the room
- One economist in on the phone

Committee votes to go down to 75%
Market crashes

12 years later: Head of foundation tells me that if Thaler had been the economist in the room, they would have saved millions.

Why is Richard Thaler persuasive?

From a 2012 conversation with the president of one of the oldest U.S. foundations working to improve social policy:



- "Uses humor"
- "Gives the impression we're all in this together"
- "Says this is something different"
- "Doesn't try to do too much"

The big picture…

The Golden Rule?

"Do unto others as you would have them do unto you."

A good way to live, but a terrible way to write, talk and present. And a common mistake.

Instead: Do unto them as they want to be done unto! Who is your audience? Tailor accordingly.

Solving the Persuasion Puzzle



@DSTILDCO

So audiences are different, but do they share any general qualities?

2 "Systems" in Every Audience Member

1) System 2: Deliberative (calculated, rational)
   - Requires arguments and evidence

2) System 1: Intuitive (automatic, emotional)
   - Requires ideas that are easily accessible and appealing

Often must satisfy both to close the "sale".

"A person usually has two reasons for doing something: a good reason and the real reason."

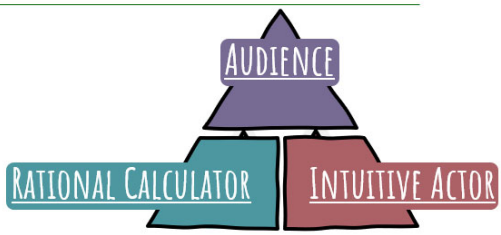
So, figure out what they need to hear (System 2), and what they want to hear (System 1).

Solving the Persuasion Puzzle



@dstildco

Solving the Persuasion Puzzle

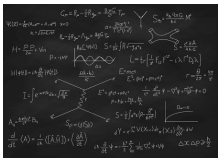


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Now what tools do you have to work with?

A framework…

Aristotle's Rhetoric



Logos
The logic of the idea

Why does your recommendation follow from the evidence?



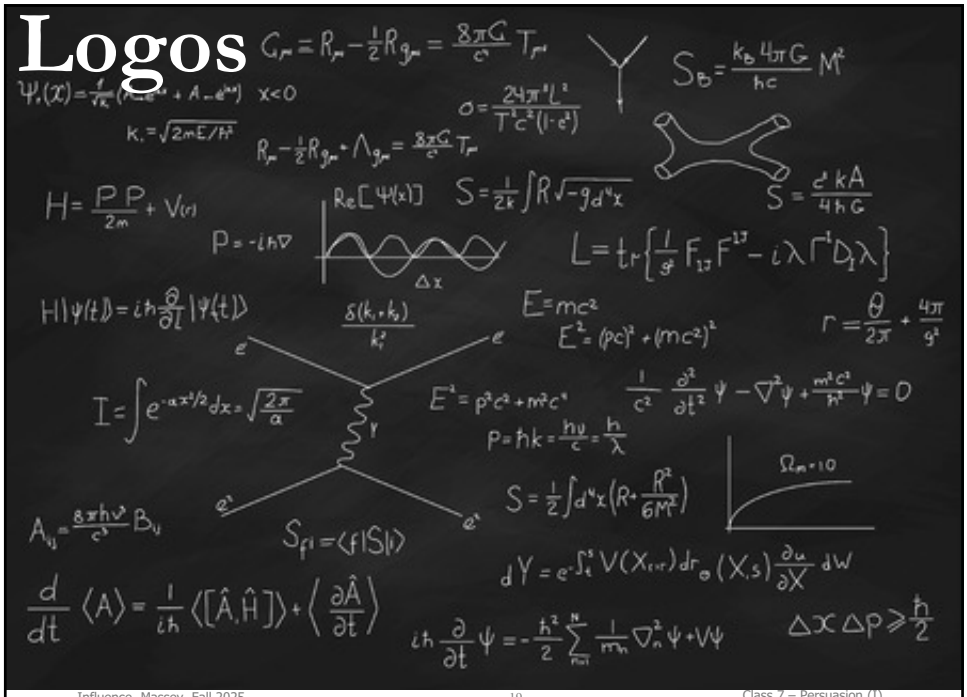
Pathos
The emotions of the audience

How can you make your proposal vivid?



Ethos
The character of the speaker

Why should the audience trust you?



Logos

Why your idea follows from the facts of the situation.

Must go beyond opinion, regardless of how strongly held. Where is your evidence?

- Data
- Research
- Logic

Logos Rx #1



Rx #1: Tailor evidence to the audience

Remember our persuasion variation on the Golden Rule: Do unto others as they want to be done unto!

What kind of evidence does your audience traffic in? E.g.,

- Doctors: studies
- Academics: p-values
- Finance: historical returns
- Football: film, or data, depending!

Example: Janette Sadik-Kahn

NYC Transportation Commissioner, 2007-2013

- >400 miles of bike lanes & 60 pedestrian plazas
- **Bloomberg administration => Data!**



Logos Rx #2



@DSTILDCO

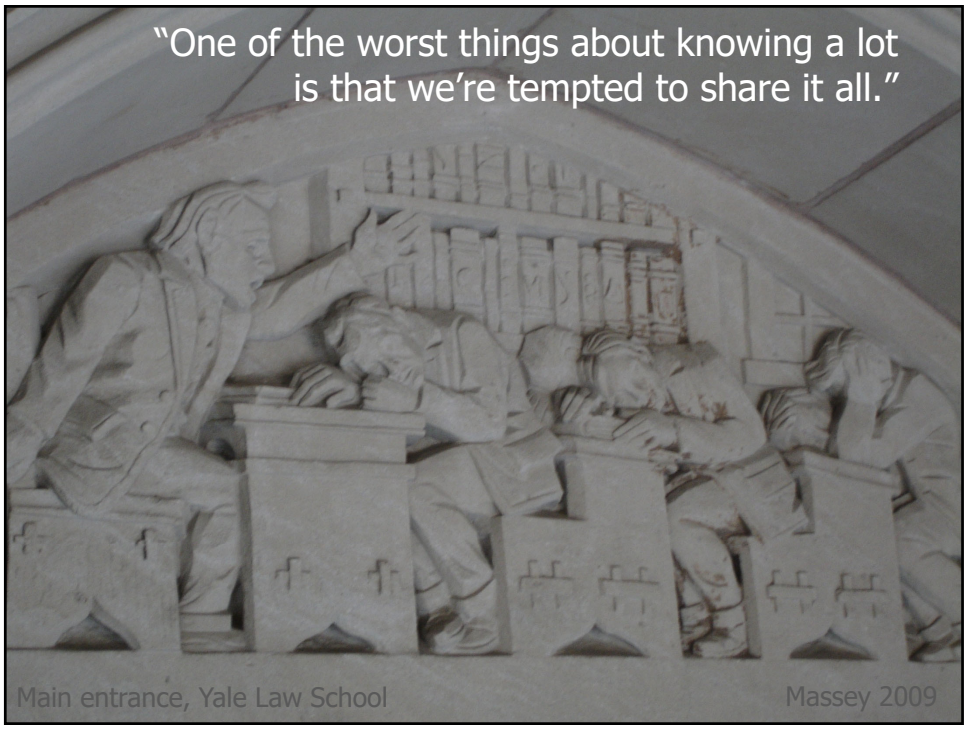
Logos Rx #2



Rx #2: Use less than you're inclined to

For most audiences, logos is a categorical requirement – they believe you have it or you don't. Make the case and move on.

The "fundamental speaker error"
Trying to present too much.



Article

People systematically overlook subtractive changes

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Published online: 7 April 2021

Check for updates

Gabrielle S. Adams[1,4], Benjamin A. Converse[1,2,4], Andrew H. Hales[1,4] & Leidy E. Klotz[3,4]

Improving objects, ideas or situations—whether a designer seeks to advance technology, a writer seeks to strengthen an argument or a manager seeks to encourage desired behaviour—requires a mental search for possible changes[1–3]. We investigated whether people are as likely to consider changes that subtract components from an object, idea or situation as they are to consider changes that add new components. People typically consider a limited number of promising ideas in order to manage the cognitive burden of searching through all possible ideas, but this can lead them to accept adequate solutions without considering potentially superior alternatives[4–10]. Here we show that people systematically default to searching for additive transformations, and consequently overlook subtractive transformations. Across eight experiments, participants were less likely to identify advantageous subtractive changes when the task did not (versus did) cue them to consider subtraction, when they had only one opportunity (versus several) to recognize the shortcomings of an additive search strategy or when they were under a higher (versus lower) cognitive load. Defaulting to searches for additive changes may be one reason that people struggle to mitigate overburdened schedules[11], institutional red tape[12] and damaging effects on the planet[13,14].

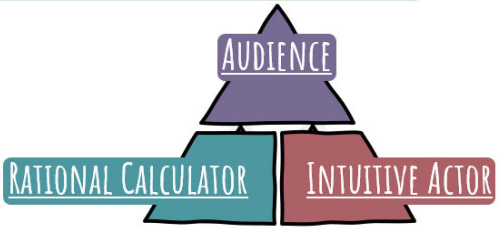
Analogous to Coco Chanel's advice

Re accessories:

"Before you leave the house, look in the mirror and take one thing off."

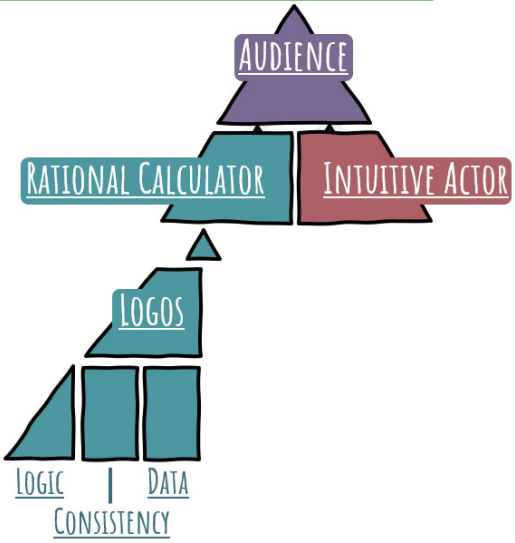


Solving the Persuasion Puzzle



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Solving the Persuasion Puzzle



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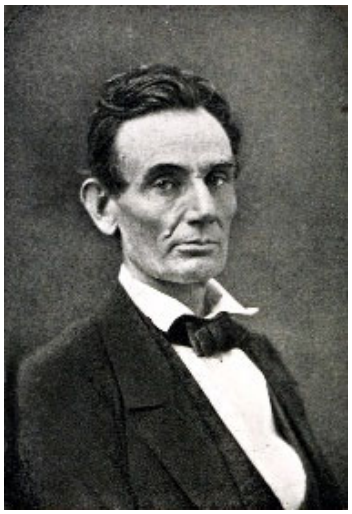
Pathos



Pathos

Present in a way that inspires emotion

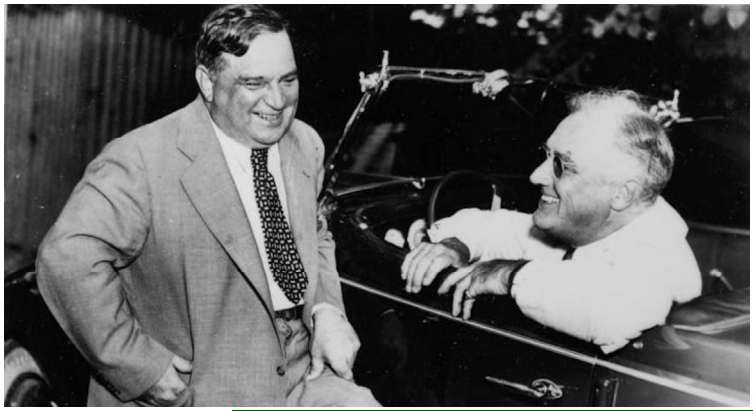
Decisions depend on more than reason alone



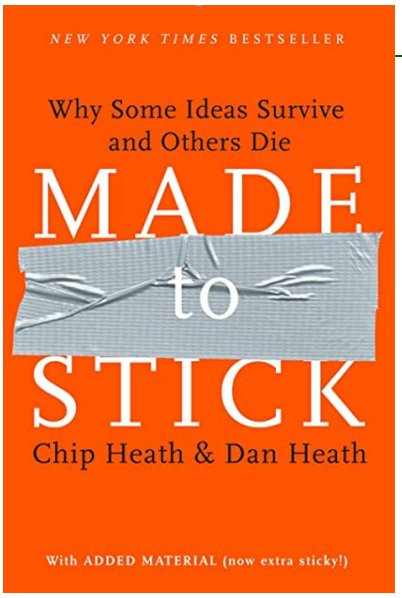
In order to "win a man to your cause" you must first reach his heart "the great high road to his reason."

FDR on Laguardia (in class-3 video)

"Our mayor is the most appealing man I know. He comes to Washington and tells me a sad story. The tears run down my cheeks, and the tears run down his cheeks. And the first thing I know, he's wangled another $50 million out of me." - FDR



Okay but how do we do that?

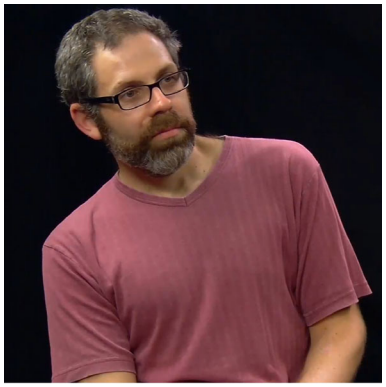


The "Telling Others" Stage

Getting a message across has two stages: The Answer stage and the Telling Others stage.

"[People] seem to believe that, once they've clicked through a PowerPoint presentation showcasing their conclusions, they've successfully communicated their idea. What they've done is shared data."

The "Telling Others" Stage



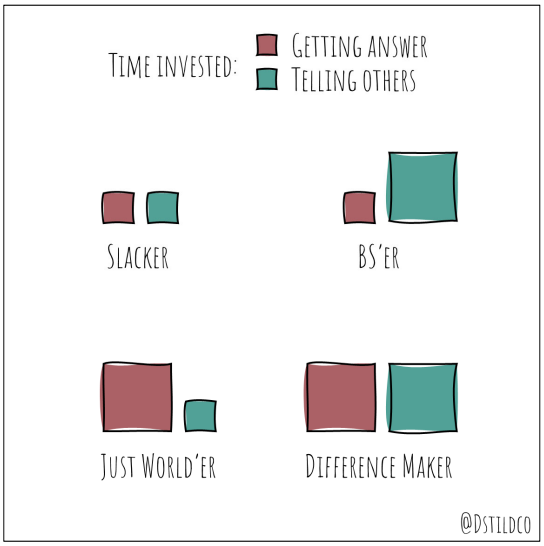
Uri Simonsohn
Co-author, "False Positive Psychology"
Co-founder, Data Colada

Spends "weeks or even months" looking for the right metaphor for a research(!) talk.



Dean Oliver
Data scientist, author
NBA executive (and coach!)

"40% of my time" is figuring out how to best communicate ideas



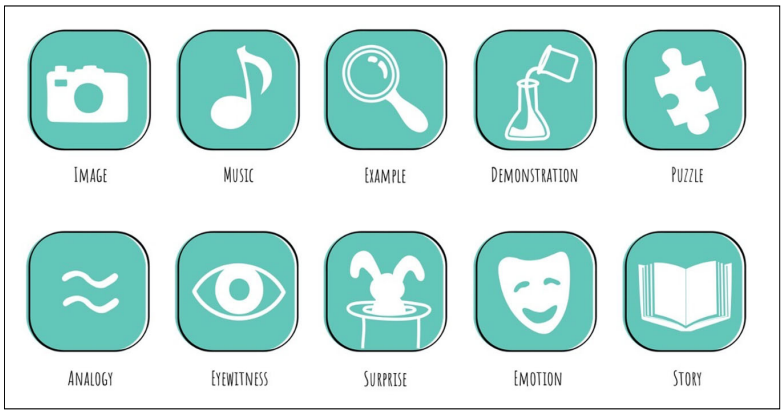
Where's your diorama?



"It never ceases to amaze me how you can talk and talk and talk to some guy about something you've got in mind, and he isn't very impressed, and then you bring in a beautiful picture of it, or better yet, a scale model with the bridge all in white and the water nice and blue, see, and you can see his eyes light up."

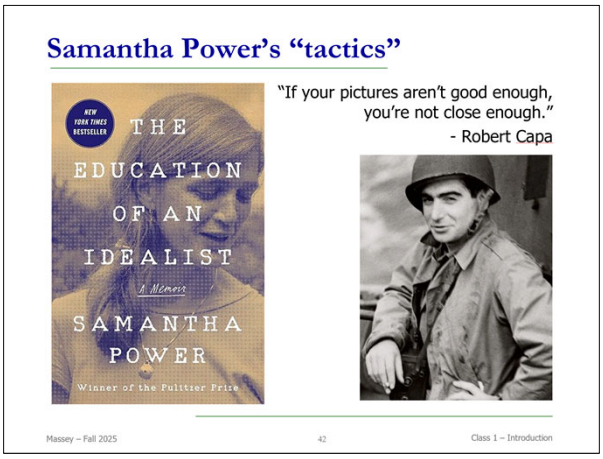
- Jack Madigan, assistant to Robert Moses

The Pathos Portfolio

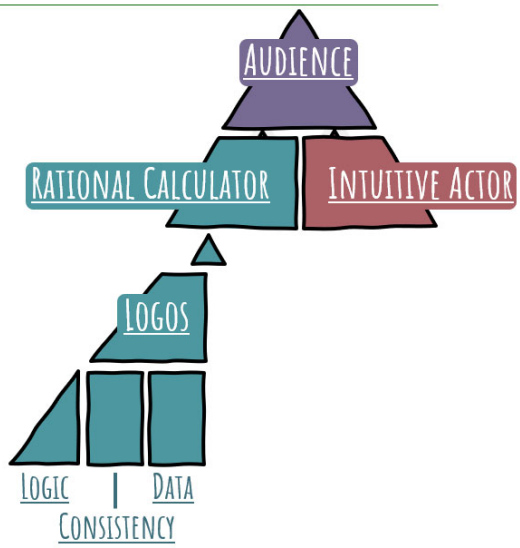


E.g., eyewitness accounts

Remember this slide from Class 1?

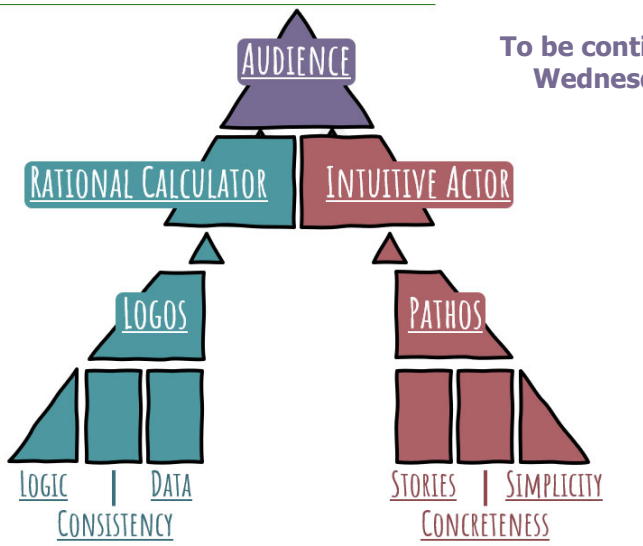


Solving the Persuasion Puzzle



@DSTILDCO

Solving the Persuasion Puzzle

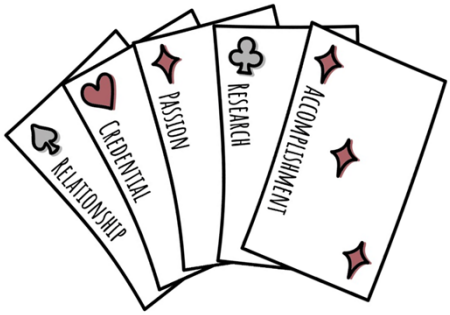


To be continued Wednesday…

Persuasive Emails

My email Rx

1) Be audience-specific
2) Use <u>your</u> voice
3) Lead with your best card
4) Shorter



Appendix

George Mayerle eye test chart

National Library of Medicine
C1907
http://resource.nlm.nih.gov/101573438

Abstract: George Mayerle's eye test chart ... "combined four subjective tests done during an eye examination. Running through the middle of the chart, the seven vertical panels test for acuity of vision with characters in the Roman alphabet (for English, German, and other European readers) and also in Japanese, Chinese, Russian, and Hebrew. A panel in the center replaces the alphabetic characters with symbols for children and adults who were illiterate or who could not read any of the other writing systems offered. Directly above the center panel is a version of the radiant dial that tests for astigmatism. On either side of that are lines that test the muscular strength of the eyes. Finally, across the bottom, boxes test for color vision, a feature intended especially (according to one advertisement) for those working on railroads and steamboats."-- Hidden treasure.

In short: Make it vivid

How one hospital finally got doctors to wash their hands more regularly:

Cultured their hands at a meeting and posted the pictures on monitors around the facility.



Made to Stick

Humans are hard-wired for stories

Stats vs. Stories

- 1-minute student speeches
- Used: 2.5 stats and 0.1 stories per student
- Remembered: 63% remember a story, 5% remember a stat

"The key is the story. A lot of people want to boil it down to a number. It's never just a number, it's a story."

- Ken Catanella, Asst VP, Sacramento Kings, Fuqua MBA 2005

